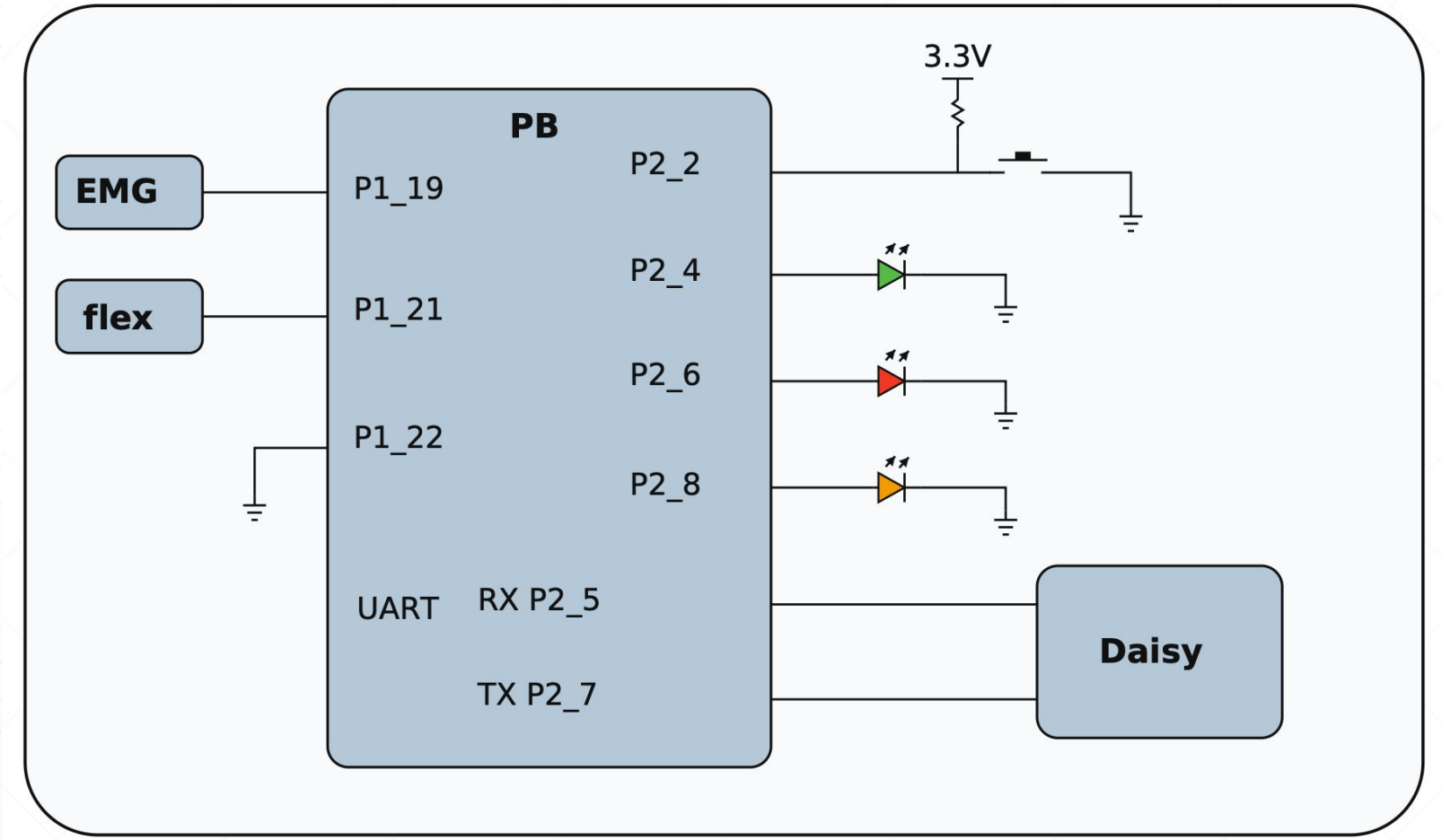
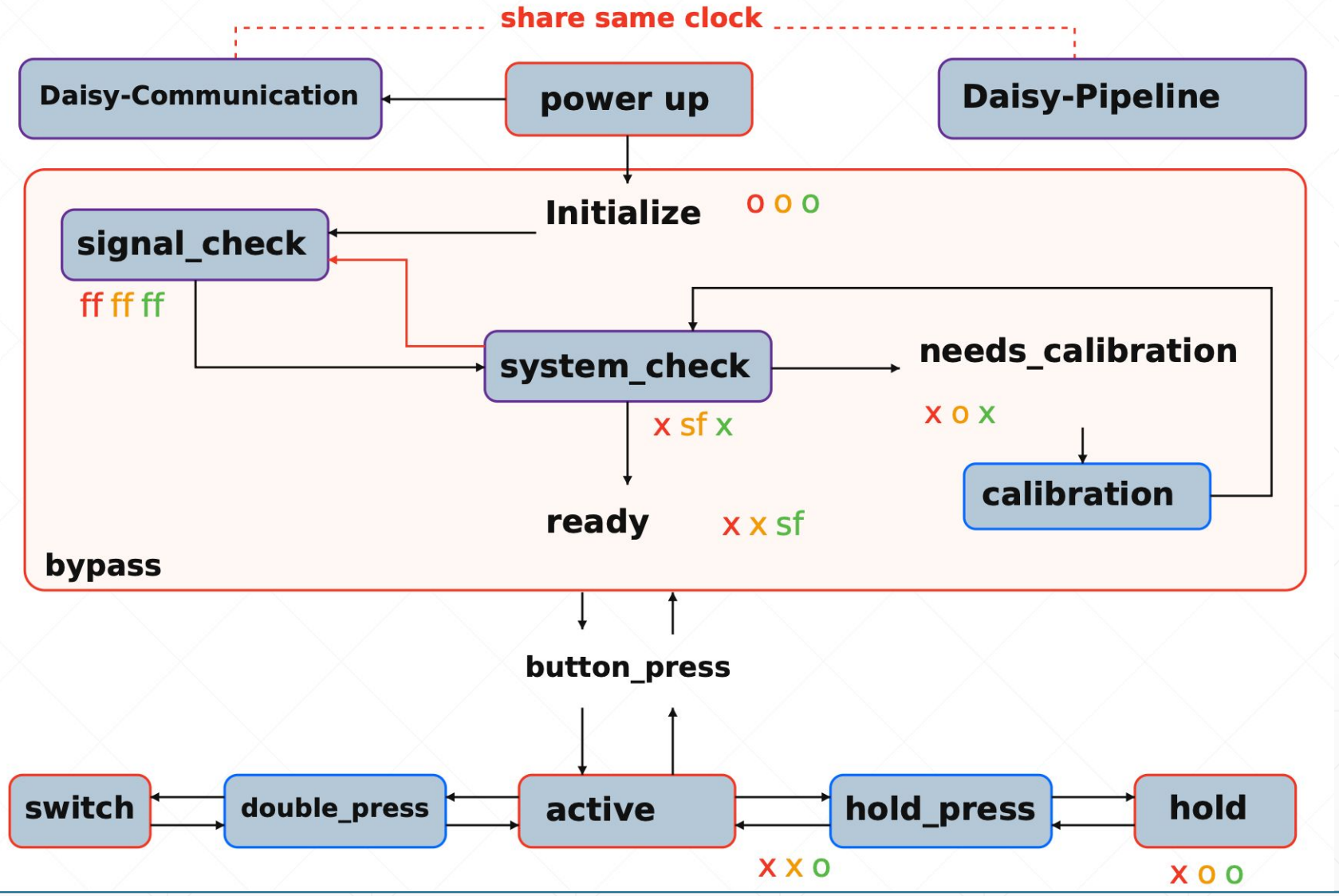
Pocket Beagle Pins Wiring



Software Flow Diagram



Function Details

calibration:

wait_for_press x o x

↓

long_press clear calibration cache

↓

3s rest x sf x

record rest EMG baseline

stretch baseline

↓

1s trans x o x

stop recording

separate data

↓

3s active x ff x

record tight EMG

flex

↓

3s validation x o x

check whether EMG can be classified

check flex range

pass failed (return value)

sf x x x x sf

system-check:

if good or failed

→ needs_calibration

if pass → ready

if bad → signal_check

signal_check

check signal quality

return good / bad

active

calculate [parameters]

by flex & EMG

every 0.2s

hold_press:

if press_time ∈ [1,4)

→ hold

if press_time > 4

→ active

Daisy-Communication:

if (status == bypass)

send bypass to Daisy

if (status == active)

send active to Daisy

update [parameters] to Daisy

if (status == hold)

fix the last [parameters] to Daisy

if (status == switch)

send switch to Daisy

update [parameters] to Daisy

if (not_com or bad_com)

display o o sf

Daisy-Pipeline:

receive and return status to PB

if receive not readable

do bypass

return bad_com

if receive == None

do bypass

return not_com

if receive == active or hold

do algorithm

if receive == switch

switch algorithm

if receive == bypass

do bypass

led status:

x : red off

sf : yellow slow flash

ff : yellow fast flash

o : green on